PRAYER

Dear Lord and Father of all, Thank you for today. Thank you for ways in which you provide for us all. For Your protection and love we thank you. Help us to focus our hearts and minds now on what we are about to learn. Inspire us by Your Holy Spirit as we listen and write. Guide us by your eternal light as we discover more about the world around us.

We ask all this in the name of Jesus.

Amen.

MATATAG CURRICULUM

TECHNOLOGY AND LIVELIHOOD EDUCATION

QUARTER 4

Short Review

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.

CARPENTER

Short Review

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.

1

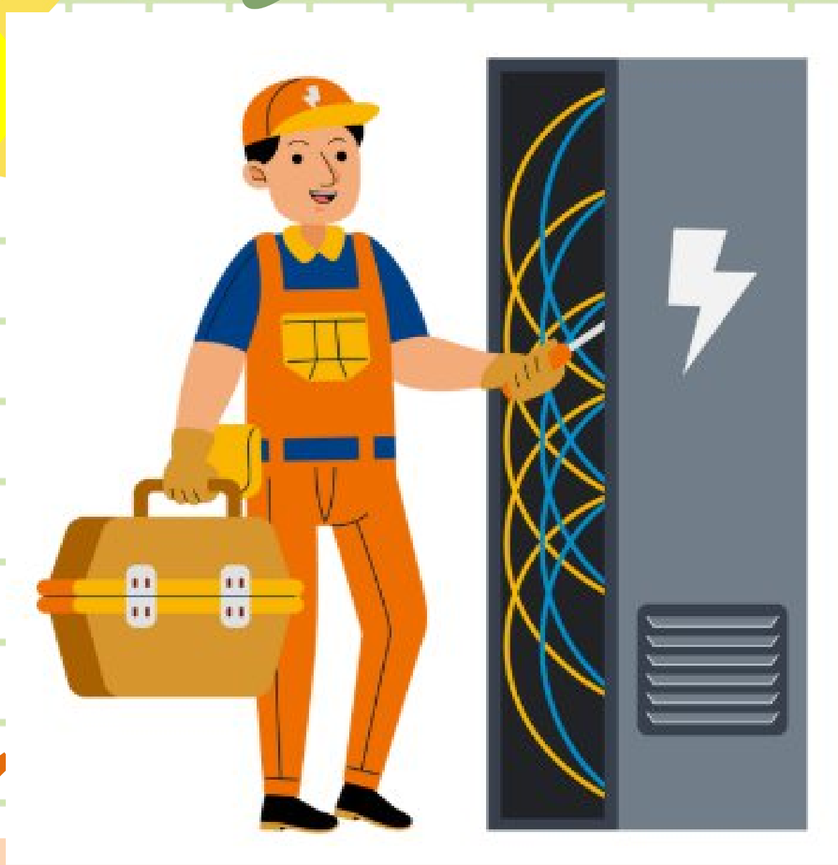
CARPENTER

Short Review

2

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.



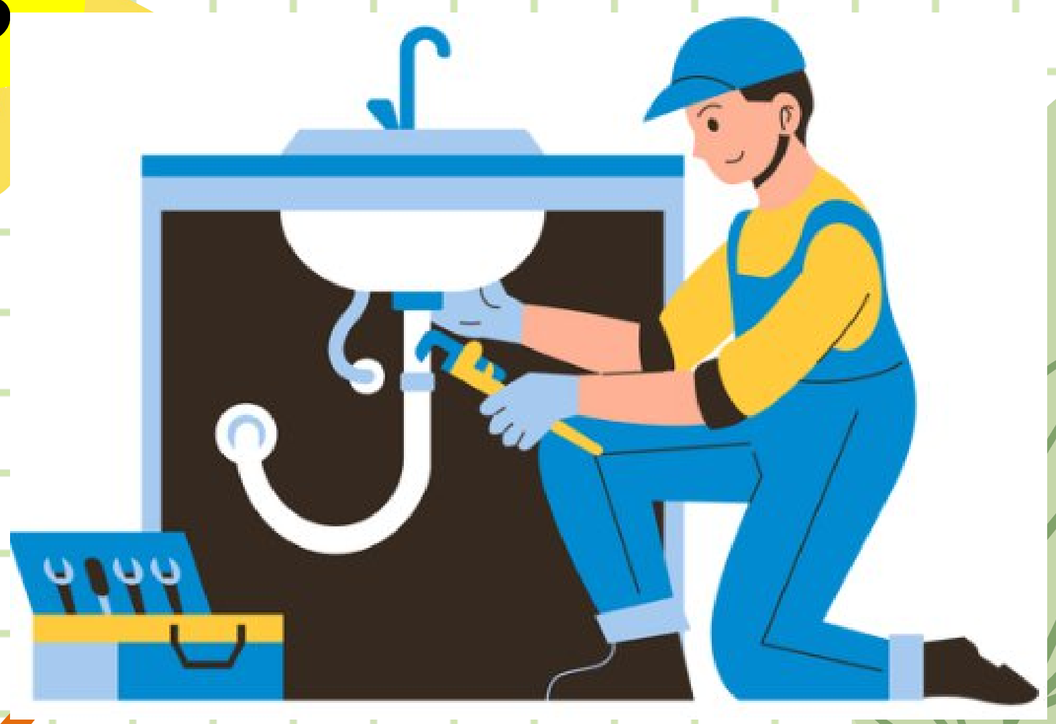
ELECTRICIAN

Short Review

3

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.



PLUMBER

Short Review

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.

AUTOMOTIVE MECHANIC

Short Review

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.

5

WELDER

Short Review

Community Builders

Directions: Identify the different careers or business opportunities in industrial arts being shown in the picture.

6

MASON

Lesson Purpose

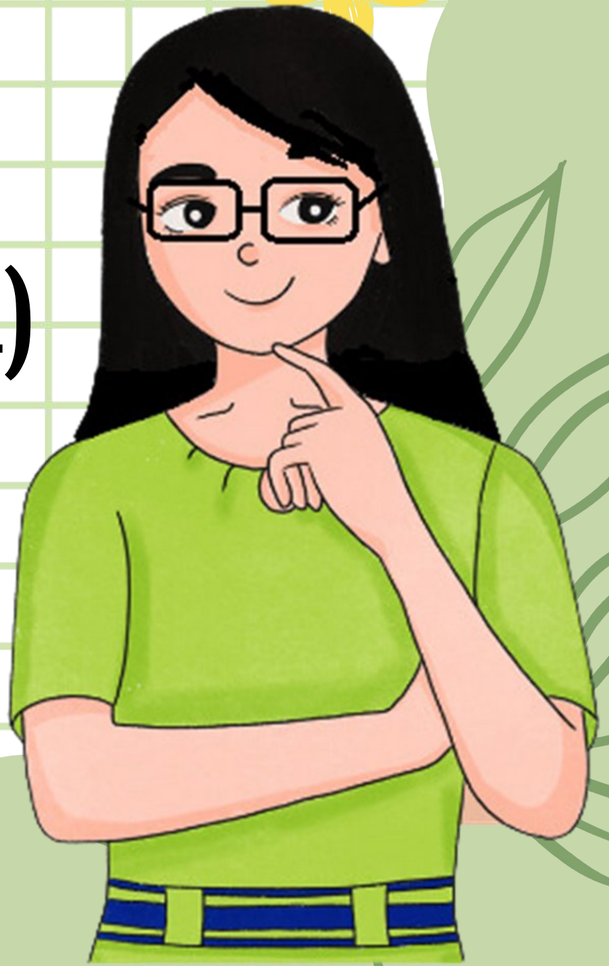
What is the picture all about?

Lesson Purpose

What do you think is our topic for today?

National Building Code of the Philippines or Republic Act No 6451 (Residential Construction)

Quarter 4 Week 2 – Day 1



Content Standards

The learners demonstrate an understanding of the concepts in industrial arts services.

Performance Standards

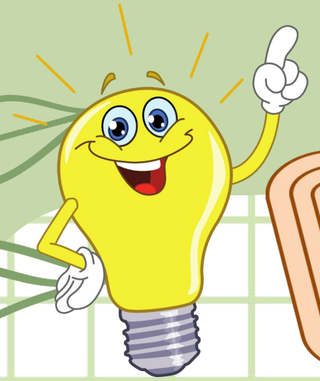
The learners perform mensuration and calculations following safety precautions.

Learning Competencies

- Discuss the basic concepts of the building code, electrical code, electronics code, traffic, and road signs.
- Explain the importance of following the set codes and standards for industrial arts services.

Learning Objectives

1. Identify the legal bases behind building code in the Philippine setting.
2. Discuss basic requirements and standards for building code of the Philippines.
3. Adhere to the importance of following the set codes and standards for building codes in daily life.



BUILDING

a structure with a roof and walls, such as a house, school, store, or factory.

REPUBLIC ACT

- is a piece of legislation used to create policy in order to carry out the principles of the Constitution. It is crafted and passed by the Congress of the Philippines and approved by the President of Philippines. It can only be repealed by a similar act of Congress.

Do you know the story of the three little pigs?

Watch video on the next slide.

Processing Questions

Who are the main characters and what are their qualities?

The three little pigs. 1st pig lazy, 2nd pig-stubborn, 3rd pig-laborious and intelligent.

Processing Questions

What are their houses built upon?

1st pig-straw, 2nd pig-wood, 3rd pig-bricks.

Processing Questions

What happened to the main characters when confronted by the wolf?

Pigs 1 and 2's houses were destroyed and went to Pig 3 for help and shelter.

Processing Questions

Can you relate the story in real life? In what area/s can you compare it?

Explicitation

1. What is the National Building Code of the Philippines?
2. What government agency/ies is responsible for implementing and regulating Republic Act 6451 or PD 1096?
3. What are the basic requirements in residential constructions?

LET'S FIND OUT!

THE NATIONAL BUILDING CODE OF THE PHILIPPINES - PRESIDENTIAL DECREE NO. 1096 Adopting A National Building Code of The Philippines Thereby Revising Republic Act Numbered Sixty-Five Hundred Forty-One (RA 6541)

Processing Questions:

1. What are the key structural requirements for residential buildings ensure safety and stability?

Key Structural Requirements for Residential Buildings:

- The National Building Code mandates that residential buildings must be designed to withstand various loads, including dead loads (weight of the structure itself), live loads (occupants, furniture, etc.), wind loads, and seismic loads.

Processing Questions:

1. What are the key structural requirements for residential buildings ensure safety and stability?

- Proper foundation, framing, and roofing systems must be used, ensuring materials comply with strength and durability standards.
- Reinforced concrete, steel, and other approved construction materials must be used as specified in the code.

Processing Questions:

2. How does the National Building Code regulate fire safety measures in residential buildings, and what are the essential fire protection systems required?

Fire Safety Measures in Residential Buildings:

- Fire-resistant materials should be used in construction, especially for walls, ceilings, and partitions.
- Residential buildings must have adequate fire exits, emergency escape routes, and access to firefighting equipment (e.g., fire extinguishers).
- Electrical and gas installations must comply with fire safety regulations to prevent fire hazards.

Processing Questions:

3. What are the minimum accessibility standards for residential buildings to accommodate persons with disabilities, as stated in the Philippine Building Code?

Minimum Accessibility Standards for Residential Buildings:

- Residences intended for public housing or multi-unit dwellings must have barrier-free features, including ramps, wide doorways, and accessible restrooms.
- Elevators and handrails should be installed in multi-story residential buildings.
- Pathways, doorways, and living spaces should be designed for wheelchair accessibility as per BP 344 (Accessibility Law).

Processing Questions:

4. How do sanitation and health standards influence the design and maintenance of residential water supply and sewage disposal systems?

Sanitation and Health Standards in Residential Construction:

- Residential buildings must have proper plumbing systems to ensure a clean and continuous water supply.
- Sewage disposal must be properly managed using septic tanks or a connection to a sewerage system to prevent contamination.
- Waste disposal and drainage systems should be properly designed to prevent flooding, leaks, and sanitation-related health hazards.

Processing Questions:

5. What environmental sustainability guidelines should be considered in residential construction to promote energy efficiency and reduce environmental impact?

Environmental Sustainability Guidelines in Residential Construction:

- The code encourages the use of energy-efficient materials and appliances, such as LED lighting and insulated walls for cooling efficiency.
- Water conservation systems, such as rainwater harvesting and low-flow plumbing fixtures, should be incorporated.
- Sustainable materials (e.g., bamboo, recycled wood) and solar panels are recommended to reduce environmental impact.

LESSON ACTIVITY:

Inspector Engineer

Let students imagine that they are the city building official and their classroom as a place of dwelling. Let them inspect through a checklist if the classroom will pass the minimum requirement for residential dwellings. 9-10 checks will mark the building passed.

LESSON ACTIVITY:

✓or X	Minimum Requirements
	Dwelling location and lot occupancy
	Light and ventilation
	Sanitation
	Foundation
	Post
	Floor
	Roof
	Stairs
	Entrance and Exit
	Electrical Requirements
	Mechanical Requirements

Explicitation

1. What is the National Building Code of the Philippines?

PD 1096 sets minimum standards for the design, construction, occupancy, and maintenance of buildings to ensure safety, accessibility, and sustainability.

Explicitation

2. What government agency/ies is responsible for implementing and regulating Republic Act 6451 or PD 1096?

The DPWH enforces the code, while LGUs issue permits and conduct inspections.

Explicitation

3. What are the basic requirements in residential constructions?

Structural integrity, fire safety, accessibility, sanitation, electrical systems, and environmental sustainability.

3-2-1 Game

3 Things I learned.

_____.

2 New words I encounter

_____.

1 Question I have in mind

_____.

Reflection on Learning

Watch the video on YouTube entitled, Understand Goal 11: Sustainable Cities and Communities (Secondary).

Write reflection towards the video in your TLE notebook.

SUSTAINABLE CITIES
AND COMMUNITIES
11
11 SUSTAINABLE CITIES
AND COMMUNITIES

Formative Assessment

Multiple-Choice

Instructions: Choose the correct answer to each question.

Formative Assessment

1. What government policy or law states that purpose of this Code is to provide for all buildings and structures, a framework of minimum standards and requirements to regulate and control their location, site, design, quality of materials, construction, use, occupancy, and maintenance?

a. Fire Code of the Philippines

b. National Building Code

c. Philippine Electrical Code

d. Philippine Electronics Code

Formative Assessment

2. What is the primary purpose of the National Building Code of the Philippines?

a. To regulate the cost of construction materials

b. To provide a framework of minimum standards for building safety and quality

c. To limit the number of buildings in urban areas

d. To encourage the use of modern architectural styles

Formative Assessment

3. Which of the following aspects is NOT specifically covered under the National Building Code for residential construction?

a. Structural design and requirements

b. Fire protection and safety

c. Accessibility and barrier-free design

d. Land ownership laws

Formative Assessment

4. Which among the following is a group of minimum requirements for a Building Permit?

a. Sanitation, Roof, Electrical Requirements, Ventilation

b. Cleanliness, Building, Mechanical Inspection, Foundation

c. Floor, Entrance and Exit, Air, History

d. Light and Ventilation, Post, Type of Persons Occupying, Address

Formative Assessment

5. What is one of the fire safety requirements for residential buildings mentioned in the National Building Code?

a. Fire exits should only be installed in high-rise buildings

b. All buildings must have fire-resistant materials, emergency exits, and access for firefighting equipment

c. Fire alarms are only required for commercial buildings

d. Residential structures are exempt from fire safety requirements

Key to Correction

1. B
2. B
3. D
4. A
5. B

Thank You